

MAT31-3

Information Theory & Statistical Learning Foundations

Convergence theory and the information-theoretic toolkit for estimation and learning

From power series and uniform convergence to entropy, Fisher information, and cross-entropy loss.

Albert School — 38 hours

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1 Power series

Definition 1 A **power series** centred at 0 is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n.$$

There is a **radius of convergence** $R \in [0, +\infty]$ such that the series converges absolutely for $|x| < R$ and diverges for $|x| > R$. The interval $(-R, R)$ is the **interval of convergence** (behaviour at $x = \pm R$ must be examined separately).

Proposition 2 (Ratio and root tests) For the power series $\sum a_n x^n$,

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \quad (\text{ratio test}), \quad \frac{1}{R} = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} \quad (\text{root test}),$$

whenever the limit exists, with the conventions $1/0 = +\infty$ and $1/(+\infty) = 0$.

Worked example. For $\sum_{n \geq 1} x^n/n$, $|a_{n+1}/a_n| = n/(n+1) \rightarrow 1$, so $R = 1$. For $\sum x^n/n!$, $|a_{n+1}/a_n| = 1/(n+1) \rightarrow 0$, so $R = +\infty$.

Theorem 3 (Term-by-term differentiation and integration) Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ have radius of convergence $R > 0$. Then f is differentiable on $(-R, R)$ and may be differentiated and integrated term by term, each resulting series having the same radius R :

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad \int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}.$$

2 Pointwise and uniform convergence

Definition 4 A sequence of functions f_n on a set D **converges pointwise** to f if for every fixed $x \in D$, $f_{n(x)} \rightarrow f(x)$. It **converges uniformly** to f if

$$\sup_{x \in D} |f_{n(x)} - f(x)| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

This is the **supremum criterion**: control of the error must be uniform in x .

Proposition 5 Uniform convergence implies pointwise convergence. The converse is false.

Worked example — Pointwise but not uniform. On $D = [0, 1]$ let $f_{n(x)} = x^n$. For each fixed x , $f_{n(x)} \rightarrow f(x)$ where $f(x) = 0$ for $x < 1$ and $f(1) = 1$, so $f_n \rightarrow f$ pointwise. But

$$\sup_{x \in [0, 1]} |f_{n(x)} - f(x)| = \sup_{x \in [0, 1]} x^n = 1 \not\rightarrow 0,$$

so the convergence is **not** uniform.

Theorem 6 (Preservation of continuity) If each f_n is continuous on D and $f_n \rightarrow f$ uniformly on D , then f is continuous on D .

Remark The example $f_{n(x)} = x^n$ illustrates the theorem's necessity: each f_n is continuous, yet the pointwise limit f is discontinuous at $x = 1$ — consistent with the convergence being only pointwise, not uniform.

3 Interchange theorems

Uniform convergence licenses the exchange of a limit with an integral or a derivative; these are theorem-governed operations, not automatic.

Theorem 7 (Limit and integral) If $f_n \rightarrow f$ uniformly on $[a, b]$ and each f_n is continuous, then

$$\lim_{n \rightarrow \infty} \int_a^b f_{n(x)} \, dx = \int_a^b \lim_{n \rightarrow \infty} f_{n(x)} \, dx = \int_a^b f(x) \, dx.$$

Theorem 8 (Limit and derivative) If $f_n \rightarrow f$ pointwise on $[a, b]$, each f_n is continuously differentiable, and the derivatives $f_{n'}$ converge uniformly to some function g , then f is differentiable and

$$f'(x) = \lim_{n \rightarrow \infty} f_{n'}(x) = g(x).$$

Common pitfall Pointwise convergence alone is not enough. With $f_{n(x)} = \sin(nx)/\sqrt{n}$ on $[0, 2\pi]$, $f_n \rightarrow 0$ uniformly, yet $f_{n'}(x) = \sqrt{n} \cos(nx)$ does not converge: limit and derivative cannot be interchanged here.

4 Fixed-point arguments

Definition 9 Let $(E, \|\cdot\|)$ be a complete normed space. A map $T : E \rightarrow E$ is a **contraction** if there is a constant $\gamma \in [0, 1)$ with

$$\|T(x) - T(y)\| \leq \gamma \|x - y\| \quad \text{for all } x, y \in E.$$

Theorem 10 (Banach fixed-point theorem) A contraction T on a complete normed space has a **unique** fixed point x^* (a point with $T(x^*) = x^*$), and for any starting point x_0 the iterates $x_{k+1} = T(x_k)$ satisfy

$$\|x_k - x^*\| \leq \gamma^k \|x_0 - x^*\| \rightarrow 0.$$

4.1 Application: value iteration on a finite discounted MDP

Definition 11 A **finite discounted MDP** has a finite state space S , finite action sets, a transition kernel $P(s' \mid s, a)$, a reward $r(s, a)$, and a discount factor $\gamma \in [0, 1)$. The **Bellman optimality operator** T acts on a value function $V : S \rightarrow \mathbb{R}$ by

$$(TV)(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V(s') \right].$$

Value functions are compared in the **sup norm** $\|V\|_\infty = \max_s |V(s)|$.

Proposition 12 (The Bellman operator is a γ -contraction) For any value functions V, W ,

$$\|TV - TW\|_\infty \leq \gamma \|V - W\|_\infty.$$

Proof. Fix a state s . Using $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$,

$$\begin{aligned} |(TV)(s) - (TW)(s)| &\leq \max_a \gamma \sum_{s'} P(s' \mid s, a) |V(s') - W(s')| \\ &\leq \max_a \gamma \sum_{s'} P(s' \mid s, a) \|V - W\|_\infty = \gamma \|V - W\|_\infty, \end{aligned}$$

since $\sum_{s'} P(s' \mid s, a) = 1$. Taking the maximum over s gives the result. $\square$

Corollary 13 (Convergence of value iteration) Since T is a γ -contraction in the sup norm on the complete space $\mathbb{R}^S$, Banach's theorem gives a unique fixed point V^* (the optimal value function), and value iteration $V_{k+1} = TV_k$ converges to it with

$$\|V_k - V^*\|_\infty \leq \gamma^k \|V_0 - V^*\|_\infty.$$

5 Shannon entropy

Definition 14 For a discrete random variable X with distribution $p(x)$, the **Shannon entropy** is

$$H(X) = - \sum_x p(x) \log p(x),$$

with the convention $0 \log 0 = 0$. With $\log = \log_2$ it is measured in **bits**. Entropy measures the average uncertainty of X .

Proposition 15 (Properties and maximum entropy) $H(X) \geq 0$, with equality iff X is deterministic. If X takes k values, then $H(X) \leq \log k$, with equality iff X is **uniform**. Thus the uniform distribution maximises entropy on a finite alphabet.

Definition 16 For a pair (X, Y) with joint distribution $p(x, y)$, the **joint entropy** is

$$H(X, Y) = - \sum_{x, y} p(x, y) \log p(x, y),$$

and the **conditional entropy** of X given Y is

$$H(X | Y) = - \sum_{x, y} p(x, y) \log p(x | y) = \sum_y p(y) H(X | Y = y).$$

Theorem 17 (Chain rule for entropy)

$$H(X, Y) = H(X) + H(Y | X) = H(Y) + H(X | Y).$$

6 Mutual information

Definition 18 The **mutual information** of X and Y is

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X) = \sum_{x, y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

It is symmetric and non-negative.

Theorem 19 (Information and independence) $I(X; Y) \geq 0$, with $I(X; Y) = 0$ if and only if X and Y are independent.

Proof. $I(X; Y) = \sum_{x, y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$ is the KL divergence (Section 10) of the joint distribution from the product of the marginals, hence ≥ 0 with equality iff $p(x, y) = p(x)p(y)$ for all x, y — that is, iff X and Y are independent. $\square$

Theorem 20 (Data processing inequality) If $X \rightarrow Y \rightarrow Z$ form a Markov chain (so Z depends on X only through Y), then

$$I(X; Z) \leq I(X; Y).$$

Processing Y to obtain Z cannot increase the information about X .

7 Exponential families

Definition 21 A family of distributions is an **exponential family in canonical form** if its density (or mass function) can be written

$$p(x; \eta) = h(x) \exp(\eta \cdot T(x) - A(\eta)),$$

where η is the **natural parameter**, $T(x)$ is the **sufficient statistic**, $h(x)$ is the base measure, and

$$A(\eta) = \log \int h(x) \exp(\eta \cdot T(x)) dx$$

is the **log-partition function** (the sum replaces the integral in the discrete case).

Worked example — Bernoulli as an exponential family. For $X \sim \text{Bernoulli}(p)$, $p(x) = p^x(1-p)^{1-x} = \exp\left(x \log \frac{p}{1-p} + \log(1-p)\right)$. So $T(x) = x$, the natural parameter is $\eta = \log \frac{p}{1-p}$, and $A(\eta) = \log(1 + e^\eta)$.

Theorem 22 (Moments and MLE) The log-partition function generates the moments of the sufficient statistic:

$$A'(\eta) = \mathbb{E}_\eta[T(X)], \quad A''(\eta) = \text{Var}_\eta(T(X)).$$

Given i.i.d. data $x_1, \dots, x_n$ with sample mean $\bar{T} = \frac{1}{n} \sum_{i=1}^n T(x_i)$, the maximum-likelihood estimate $\hat{\eta}$ solves the **moment-matching** equation

$$A'(\hat{\eta}) = \bar{T}.$$

Proof. The log-likelihood is $\ell(\eta) = \sum_i (\eta \cdot T(x_i) - A(\eta)) + \text{const}$. Setting $\ell'(\eta) = \sum_i T(x_i) - nA'(\eta) = 0$ gives $A'(\hat{\eta}) = \frac{1}{n} \sum_i T(x_i) = \bar{T}$. $\square$

8 Fisher information

Definition 23 For a parametric model with log-likelihood $\ell(\theta; x) = \log p(x; \theta)$, the **score function** is

$$s(\theta; x) = \frac{\partial}{\partial \theta} \log p(x; \theta).$$

Under regularity, $\mathbb{E}_\theta[s(\theta; X)] = 0$.

Definition 24 The **Fisher information** of the model at θ (for one observation) is, in two equivalent forms,

$$I(\theta) = \text{Var}_\theta(s(\theta; X)) = \mathbb{E}_\theta[s(\theta; X)^2] = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2} \log p(X; \theta)\right].$$

Proposition 25 (Equivalence of the two definitions) Differentiating $\int p(x; \theta) dx = 1$ twice under the integral sign gives $\mathbb{E}[s] = 0$ and

$$\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log p\right] + \mathbb{E}[s^2] = 0,$$

so the variance-of-score and negative-expected-Hessian forms agree.

Remark Fisher information is the **curvature** of the expected log-likelihood at θ : a sharply peaked log-likelihood (large $I(\theta)$) pins down the parameter precisely. For n i.i.d. observations the information is $nI(\theta)$.

9 Cramér–Rao lower bound

Theorem 26 (Cramér–Rao bound) For any **unbiased** estimator $\hat{\theta}$ of θ based on n i.i.d. observations,

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}.$$

No unbiased estimator can have variance below this bound.

Definition 27 An unbiased estimator is **efficient** if it attains the Cramér–Rao bound, i.e. $\text{Var}(\hat{\theta}) = 1/(nI(\theta))$.

Remark (Minimum sample size) To guarantee an unbiased estimator can achieve variance at most v , the bound $1/(nI(\theta)) \leq v$ forces

$$n \geq \frac{1}{vI(\theta)}.$$

10 Kullback–Leibler divergence

Definition 28 The **Kullback–Leibler (KL) divergence** from Q to P is

$$D_{\text{KL}}(P \parallel Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

(an integral in the continuous case). By convention the sum is restricted to x with $P(x) > 0$.

Proposition 29 (Properties) $D_{\text{KL}}(P \parallel Q) \geq 0$, with equality iff $P = Q$ (Gibbs' inequality). It is **not symmetric**: in general $D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$, and it is not a distance.

Common pitfall KL divergence is **pathological** when the supports disagree: if $Q(x) = 0$ while $P(x) > 0$ for some x , then $D_{\text{KL}}(P \parallel Q) = +\infty$. The asymmetry matters — $D_{\text{KL}}(P \parallel Q)$ penalises Q for assigning low probability where P is large, but the reverse divergence does not.

11 Cross-entropy and the link to learning

Definition 30 The **cross-entropy** of P relative to Q is

$$H(P, Q) = - \sum_x P(x) \log Q(x).$$

Proposition 31 (Decomposition)

$$D_{\text{KL}}(P \parallel Q) = H(P, Q) - H(P), \quad \text{equivalently} \quad H(P, Q) = H(P) + D_{\text{KL}}(P \parallel Q).$$

Proof. $H(P, Q) - H(P) = - \sum_x P(x) \log Q(x) + \sum_x P(x) \log P(x) = \sum_x P(x) \log \frac{P(x)}{Q(x)} = D_{\text{KL}}(P \parallel Q).$ $\square$

Theorem 32 (Minimizing cross-entropy = maximizing likelihood) Let P be the empirical distribution of data $x_1, \dots, x_n$ and let Q_θ be a model. Then

$$H(P, Q_\theta) = -\frac{1}{n} \sum_{i=1}^n \log Q_\theta(x_i) = -\frac{1}{n} \ell(\theta),$$

so minimizing the cross-entropy loss over θ is exactly maximizing the log-likelihood $\ell(\theta)$.

Remark In neural-network training the model outputs a distribution Q_θ over classes and the target is the one-hot empirical distribution P . Since $H(P)$ does not depend on θ , the decomposition $H(P, Q_\theta) = H(P) + D_{\text{KL}}(P \parallel Q_\theta)$ shows that minimizing cross-entropy minimizes $D_{\text{KL}}(P \parallel Q_\theta)$, driving the model distribution toward the data — which is the same as maximizing likelihood.